

Pavan Hegade

AI/ML Engineer • Generative AI, LLMs & RAG Systems • Full-Stack Development

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Professional Summary

Recent B.E. graduate in Artificial Intelligence and Data Science with hands-on experience building Generative AI and LLM-based applications, Retrieval-Augmented Generation (RAG) pipelines, and REST APIs using object-oriented Python. Skilled in LangChain, LangGraph, embeddings, transformer-based models, and vector databases, with additional exposure to full-stack web development. Eager to contribute to production-level AI systems as an AI/ML Intern.

Technical Skills

- **Programming:** Python (OOP), Java, JavaScript (ES6+), TypeScript
- **AI / ML:** Generative AI, LLM Integration (OpenAI, Gemini, Groq), RAG, Embeddings, Transformer Models (Hugging Face, Sentence-Transformers), Agentic AI, LangChain, LangGraph
- **Backend & Web:** FastAPI, Flask, Django REST Framework, REST API Design, Web Architecture, React.js, Async/Await
- **Databases & Vector Stores:** MySQL, PostgreSQL, MongoDB (NoSQL), Pinecone, FAISS, Chroma
- **Cloud & DevOps:** AWS (Foundational), Docker, Git, Vercel, Render, Firebase

Projects

MLPrep AI — AI-Powered Data Analyst & ML Preparation Suite

Apr 2026 – Jun 2026

Python, FastAPI, React.js, LangGraph, LangChain, PostgreSQL, Firebase, Gemini, Groq | [Live](#) | [GitHub](#)

- Built a full-stack platform for natural-language dataset analysis (CSV, Excel, JSON, Kaggle), powered by a 7-agent LangGraph pipeline for profiling, insights, and visualization.
- Engineered a sandboxed LLM code-generation workflow with self-healing retry logic and tiered routing across Gemini and Llama 3.3 (Groq).
- Delivered interactive Vega-Lite visualizations and persistent multi-turn chat memory secured with Firebase JWT auth and PostgreSQL/Firestore.

GoAttend — Full-Stack Event Management System

Feb 2026 – May 2026

Python, Django REST Framework, PostgreSQL, HTML, CSS, JS | [Live](#) | [GitHub](#)

- Built a full-stack event management and ticketing platform with secure JWT-based role-based access control (RBAC).
- Deployed frontend on Vercel and backend on Render, integrating PostgreSQL and Cloudinary.
- Designed scalable REST APIs and optimized event-booking workflows for real-time ticket management.

Pharmaceutical AI — Multi-Agent QMS Automation System

Python, FastAPI, LangGraph, Groq, SQLAlchemy, React 19, Redux | [Demo Video](#) | [GitHub](#)

- Designed a multi-agent LLM system automating pharma QMS workflows via a 7-node LangGraph state machine (Intake, Completeness, Duplicate Detection, Risk, Root Cause, CAPA, Summary).
- Integrated Groq LLMs with automatic model failover and async SQLAlchemy ORM, paired with a real-time React 19 + Redux frontend.
- Implemented TF-IDF/cosine-similarity duplicate detection, automated FMEA risk assessment, and 5-Whys/Ishikawa root-cause analysis.

Experience

Infosys Springboard — AI Intern

Oct 2025 – Dec 2025

Remote

- Analyzed large-scale datasets with Pandas and NumPy to surface patterns and actionable insights.
- Performed geospatial analysis using Folium and Matplotlib for visualization and reporting.
- Built Tableau dashboards for interactive analytics and automated reporting workflows.
- Handled data preprocessing, cleaning, visualization, and ML-based analysis tasks end to end.

TAP Academy — Full Stack Web Development Trainee

Jan 2026 – Aug 2026

Bengaluru, India

- Completed an intensive Full Stack Web Development program covering frontend (HTML, CSS, JavaScript, React.js) and backend development (Python, Django REST Framework, FastAPI) with relational database design in SQL, PostgreSQL, and MySQL.
- Built and deployed full-stack applications implementing RESTful APIs, JWT-based authentication, and MVC architecture; practiced Git/GitHub version control, CI/CD pipelines, unit testing, debugging, and Agile/Scrum workflows.
- Selected via the institution's Campus-to-Corporate (C2C) recruitment drive for industry-oriented, project-based software development training with cloud deployment (Render, Vercel, Docker).

Education

Angadi Institute of Technology and Management

2022 – 2026

B.E. in Artificial Intelligence and Data Science — CGPA: 7.5 | Belagavi, Karnataka

Certifications

- Generative AI & LLMs Foundations: From Basics to Application — Udemy
- Google Analytics Certification — Google Cloud Skills